

Macro-Economic Blame Event Study

Do Investors Punish Firms That Blame the Economy?

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1. Economic Motivation

Each quarter, executives of publicly traded companies explain how their company performed on an earnings call. Investors listen carefully to not just to the numbers, but to *how* management explains those numbers.

When a company misses expectations, management has two choices: take responsibility for what went wrong, or point to forces outside their control e.g. inflation, supply chains, interest rates, the macro environment. We call the second option “blame-shifting.”

This matters to investors as if a firm blames the macro environment more than other firms in the same industry (who face the same macro conditions), investors may view this as a failure to take accountability. This perception leads investors to believe that underperforming areas will be ignored rather than fixed, so they update their expectations downward. As investors update their beliefs during or after the call, stock prices adjust accordingly.

Hypothesis: Firms that increase macro-economic blame more than their industry peers in a given quarter tend to experience more negative cumulative abnormal returns (CAR) that is, returns beyond what the market model predicts, in the days following their earnings calls.

A key point is that we are doing a relative comparison. Firms in the same industry should face the similar macro headwinds. If one firm changes its blame significantly more than its peers, it is not the macro conditions that explain the difference - it is the firm’s choice to lean on them.

2. Dataset

We pulled earnings call transcripts from Capital IQ via WRDS for all US-based firms from 2010 to 2024. After cleaning, this resulted in approximately 181,000 earnings call transcripts across 6,294 companies.

For each call, we matched: (a) the transcript text, (b) IBES analyst estimates to compute earnings surprise (actual EPS minus the mean analyst estimate), (c) CRSP daily stock returns around the call date, and (d) Fama-French 12-industry classifications to group firms by sector.

3. Design Choices and Justification

3.1 Estimation Window: $[-135, -15]$

The estimation window is the period we use to estimate each stock’s “normal” behavior relative to the market. We chose a 120-trading-day window ending 15 days before the earnings call ($[-135, -15]$).

We chose this range for two reasons. First, 120 days gives us enough data to estimate a reliable beta (the stock’s sensitivity to the market). Second, the 15-day gap prevents information leakage: if we estimated normal returns using data right up to the event, any pre-announcement drift or anticipation would contaminate our baseline. The gap ensures our “normal” benchmark is clean.

3.2 Event Window: [0, +2]

The event window is the period where we measure abnormal returns i.e. the market's reaction to the earnings call. We use [0, +2], meaning the day of the call through two trading days after.

Day 0 captures the immediate reaction (specifically included to avoiding missing firms with earnings calls before/during trading hours). Days +1 and +2 allow for delayed processing, which is needed because earnings calls often happen after market close, and institutional investors may take a day or two to fully digest the information. Going beyond +2 increases risks picking up noise from unrelated events.

4. Estimation Model

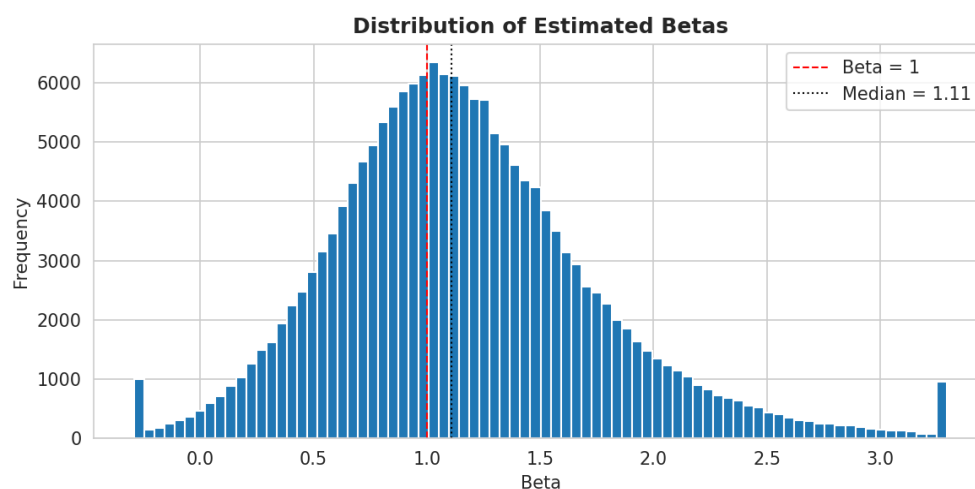
We use the market model to estimate expected returns. The market model says a stock's expected return on any given day is a linear function of the overall market's return that day:

$$E(R_i) = \alpha_i + \beta_i \cdot R_m$$

where α (alpha) is the stock's average return independent of the market, β (beta) is the stock's sensitivity to market moves, and R_m is the market return (CRSP value-weighted index).

We chose the market return model over alternatives for several reasons. The mean-return model (which just uses the stock's average return as the benchmark) ignores how much of a stock's movement is simply tracking the market. If the market rises 2% and a stock rises 2.5%, the mean model might call that "abnormal" when really the stock was just moving with the market. The market model strips out this market-wide component, isolating the firm-specific reaction.

The Fama-French 3-factor model adds size and value factors. While more complete in theory, it requires estimating three betas instead of one, which is noisier in short estimation windows. For our purposes the market model strikes the right balance between precision and simplicity.



5. Formula Specifications

The abnormal return (AR) on day t is the actual return minus the expected return from the market model:

$$AR_{i,t} = R_{i,t} - (\alpha_i + \beta_i \cdot R_{m,t})$$

The cumulative abnormal return (CAR) sums these daily abnormal returns over the event window:

$$CAR_i(0,+2) = AR_{i,0} + AR_{i,1} + AR_{i,2}$$

We use CAR rather than a single day's AR because the information from an earnings call doesn't get priced in all at once. CAR aggregates the full reaction. We also do not average across firms to get CAAR (cumulative average abnormal return) in our regression. Instead, we keep each event as its own observation. With each firm having a CAR, rather than one aggregated measure, we can obtain many datapoints, enabling us to run a regression across firms on CAR to test the significance of the blame effect.

5.1 Regression Specification

The cross-sectional regression we run each quarter is:

$$CAR_i = \gamma_0 + \gamma_1 \cdot \text{Earnings Surprise}_i + \gamma_2 \cdot \text{Blame Score}_i + \varepsilon_i$$

We include earnings surprise as a control because it is the dominant driver of CARs around earnings calls. Without it, blame-shifting might simply proxy for bad earnings news. By controlling for it, we isolate the incremental effect of language beyond the numbers themselves.

Our goal: find a statistically significant γ_2 (Blame Score coefficient) with t-stat < -2 , meaning blame-shifting predicts negative returns *above and beyond* what earnings surprise already explains.

6. Blame Score Construction

6.1 Fine-Tuning FinBERT

To create a blame score, we need to read 181,000 transcripts and score how much each one blames the macro environment. We did this in two stages using a few shot classification approach:

Stage 1: Labelling with Llama 3.2 (3B)

First, we extracted 22,000 sentences containing words from a macro-economic dictionary (terms like "inflation," "supply chain," "interest rates"). We fed each sentence to Llama 3.2, a large language model, with a carefully designed prompt asking: "Is this sentence blame-shifting (the firm attributing weak performance to external macro conditions) or not?" Llama returned a label (1 = blame, 0 = not blame) and a confidence score. We kept only sentences where Llama was at least 70% confident, giving us a high-quality training set.

Llama 3.2 Training Prompt

You are a financial analyst coding earnings call transcripts for academic research on managerial attribution bias.

Identify when management attributes their firm's weak or disappointing performance to external macroeconomic or industry-wide conditions — rather than to their own decisions.

LABEL 1 (blame-shifting) when:

- Management links weak results, volume declines, margin compression, or missed targets to external macroeconomic forces
- Industry-wide pressure is cited as the primary explanation ("everyone is seeing this", "the whole industry faced headwinds", "this is due to the economy")
- External conditions (rates, inflation, FX, consumer weakness, geopolitics) are framed as the main driver of underperformance
- Headwinds language is used to contextualize disappointing outcomes

LABEL 0 (not blame-shifting) when:

- Underperformance is explicitly attributed to a firm's OWN strategic decisions (pricing strategy, contract terms, restructuring, volume-for-price tradeoffs)
- The speaker clarifies that macro/industry trends did NOT drive the result
- Text is an analyst question, not a management response
- Macro or volume data is presented neutrally as context with no performance attribution
- Management is discussing cost actions, working capital, or internal spending changes with no external blame

CRITICAL DISTINCTIONS — these are NOT blame-shifting:

- "Our shipments were down due to contract renegotiations we chose" → firm decision, label 0
- "The decline in beer volumes reflected our price-over-volume strategy, not an industry shift" → firm explicitly rejects external blame, label 0
- "We expect to reduce spending by \$20-30 million" → internal action, no external attribution, label 0
- Reporting mixed results across segments without attributing the weak ones to macro → label 0

THESE ARE blame-shifting:

- "Everybody in the industry is seeing the same pressures" → industry-wide framing to explain poor results, label 1
- "The same things that were headwinds in Q2 largely become tailwinds in Q3" → using headwind/tailwind language to explain performance swing, label 1
- "Cost inflation pressures are being felt across the industry and we are working to pass them through" → external cost pressure cited for margin issues, label 1

Return ONLY valid JSON, no markdown, no extra text:

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{"label": 0 or 1, "confidence": 0.0-1.0, "rationale": "one sentence naming the specific signal"}
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Stage 2: Fine-tuning FinBERT

FinBERT is a smaller, finance-specific language model (110M parameters vs. Llama's 3B, making it 27x smaller). We fine-tuned FinBERT on Llama's labels via gradient descent, training it to predict the probability that any given sentence is blame-shifting. The result is a fast, lightweight model that can score every sentence in every transcript.

We did it in two stages because Llama is accurate but slow and expensive to run on 181,000 full transcripts. FinBERT is fast but needs training data. Using Llama to label a subset, then distilling that knowledge into FinBERT, gives us the best of both worlds.

6.2 Scoring Methodology

For each transcript, we compute a blame intensity: the sum of FinBERT's blame probabilities across all macro-related sentences, divided by total sentences. This gives us a number between 0 and 1 representing what fraction of the call is spent blame-shifting.

However, the speakers for each firm may have different baseline levels of economic blame shifting behaviour. As such, we want to focus on the change. To adjust for this, we compute the percent change vs. the firm's own rolling 8-quarter average. A firm that suddenly blames the macro environment much more than usual is sending a stronger signal.

Finally, we make it industry-relative. We subtract the average blame-change among same-industry peers who reported in the prior 30 days. This is critical as if every tech firm is talking about macro headwinds, that would be an industry-wide narrative, not a signal about any one firm. Only the excess blame should matter.

The final blame score formula is as follows:

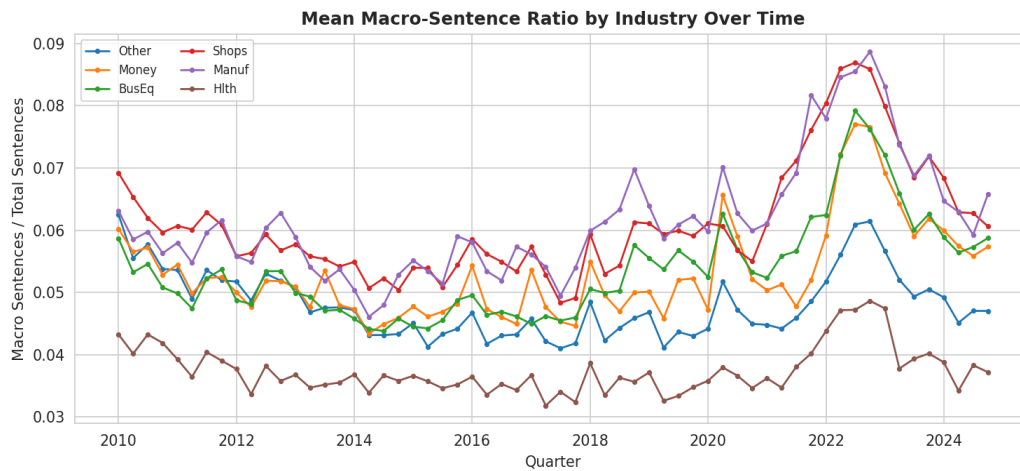
Blame Score = (Firm's blame change vs. own history) – (Average blame change among industry peers).

Firms with fewer than 3 peers in the window are excluded as they are not suitably compared.

7. Exploratory Data Analysis

Before running the event study, we profiled the data to check for issues and get an idea of what the data looks like.

The dataset spans 2010–2024 with around 181,000 calls across 12 Fama-French industries. We plotted the average macro-sentence ratio in the following industries to spot check our blame score mechanism against our knowledge of macro trends: Manufacturing, Business Equipment, Money (financials), Shops, and Health.



All industries follow a similar trend, peaking around 2020–2023 when the macro environment was most volatile (COVID, inflation, supply chain disruptions). This is reassuring: our blame proxy accurately reflects when blame-shifting should be most prevalent.

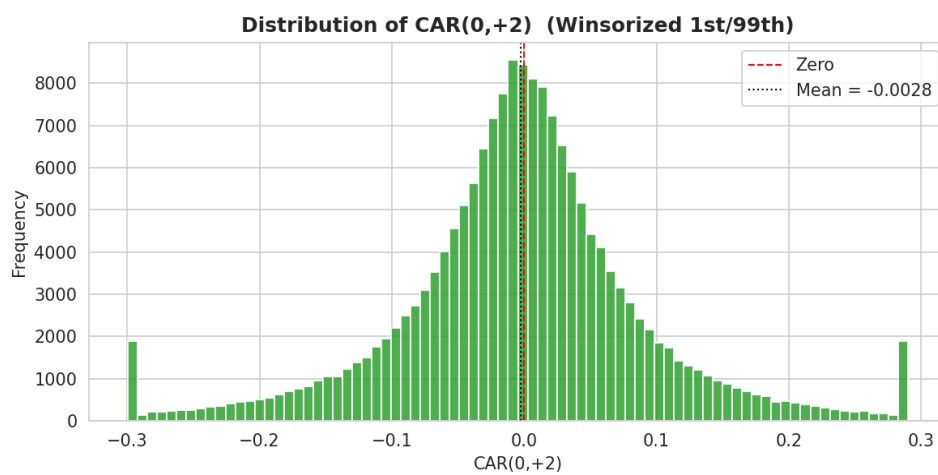


Figure 2 shows the distribution of CAR(0,+2). It is roughly symmetric around zero with mean = -0.0028 and standard deviation = 0.107.

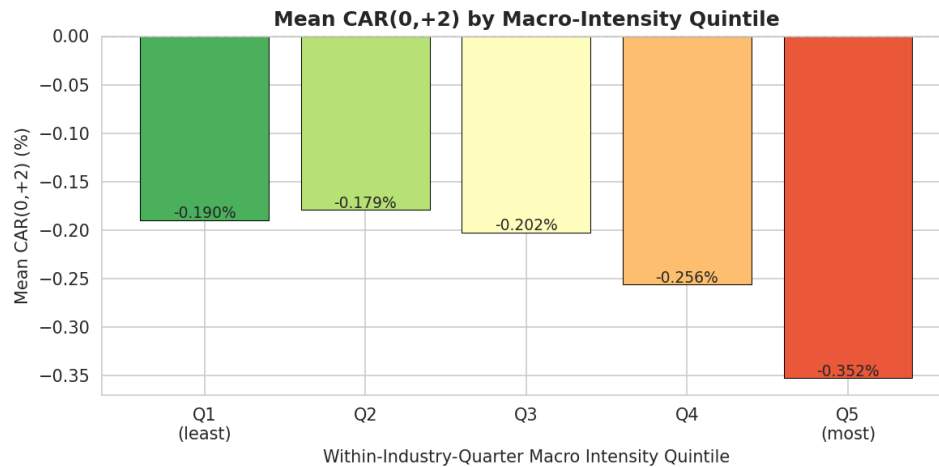


Figure 3 provides a non-parametric preview of our hypothesis. We bucket calls by within-industry-quarter macro intensity and plot mean CAR for each quintile. The pattern is suggestive: firms in the highest blame quintile (Q5) tend to have lower mean CAR than those in the lowest (Q1).

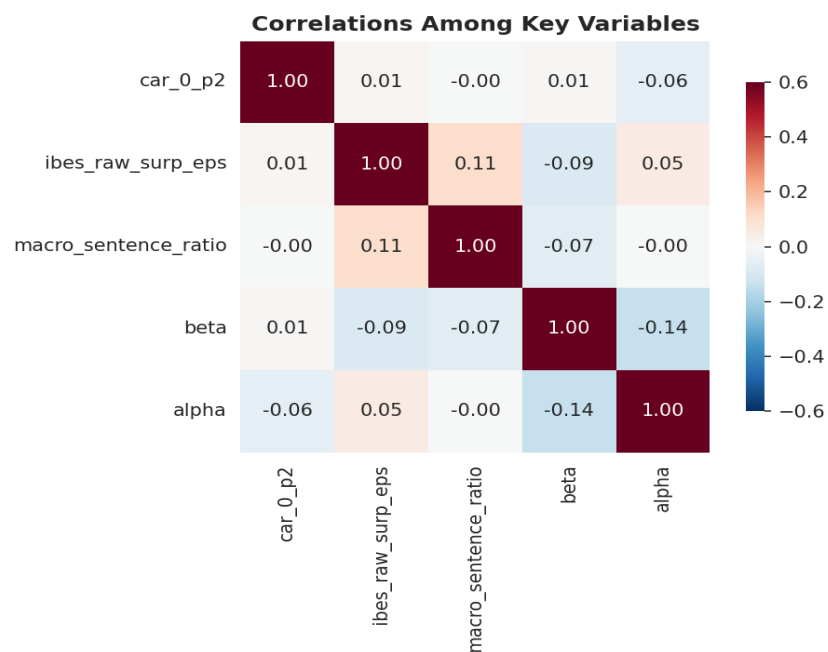


Figure 4 shows correlations among key variables. Earnings surprise has a positive correlation with CAR(0,+2), as expected. The macro-sentence ratio has a small negative correlation with CAR, consistent with the hypothesis.

8. Results

8.1 Why Fama-MacBeth

A naive approach would pool all 140,600 observations and run a single OLS regression. However, earnings calls happen in clusters (every quarter), so observations within the same quarter are not independent. Standard errors from pooled OLS would be artificially small, making everything look more significant than it really is.

The Fama-MacBeth procedure fixes this. It works in two steps:

Step 1: Run a separate cross-sectional regression for each quarter. This gives us a list of quarter-specific coefficients (γ_0 , γ_1 , γ_2 for each quarter).

Step 2: Take the time-series average of these coefficients and test whether the average is significantly different from zero using a t-test on the time series of estimates.

This approach treats each quarter as one independent observation, which is the correct unit of independence. We also apply Newey-West standard errors to account for any autocorrelation in the coefficient series across quarters.

Within each quarter, we winsorize all variables at the 1st and 99th percentiles to limit the influence of extreme outliers. Importantly, we winsorize *within* each cross-section, not globally — otherwise one quarter's tails would be trimmed by another quarter's distribution.

8.2 Fama-MacBeth Results

	Coefficient	t-stat	p-value
Intercept (γ_0)	-0.003172	-3.575	0.0003
Earnings Surprise (γ_1)	0.002578	4.567	0.0001
Blame Score (γ_2)	-0.000102	-8.512	0.0000

Table 1: Fama-MacBeth regression results. N = 140,600.

The results support the hypothesis:

Earnings surprise (γ_1) is significantly positive ($t = 4.567$). This is expected: firms that beat analyst estimates have positive abnormal returns. It confirms our regression is working as a sanity check.

Blame score (γ_2) is significantly negative ($t = -8.512$, $p < 0.0001$). Firms that shift blame to the macro environment more than their industry peers experience more negative CARs, even after controlling for earnings surprise. This is the central finding.

8.3 Economic Magnitude

Statistical significance does not mean economic significance. The blame score coefficient ($\gamma_2 = -0.000102$) is small in absolute terms. The estimated return contribution from blame (Blame Score $\times \gamma_2$) has a mean of roughly -0.02% , with an interquartile range of $[-0.3\%, +0.5\%]$.

This means blame-shifting explains a small slice of the overall CAR variation. Given the noise in daily returns (CAR std $\approx 10.7\%$), this is expected; no single qualitative signal will dominate. But the signal is real, statistically robust, and has a clear economic interpretation.

9. Conclusion and Next Steps

Hypothesis supported. Firms that increase macro-economic blame relative to their industry peers experience statistically significant negative abnormal returns after earnings calls. The effect survives controlling for earnings surprise and is robust across multiple econometric specifications.

Limited economic power. The blame signal alone produces returns typically in the $[-0.3\%, +0.5\%]$ range. This makes it meaningful as an academic finding, but insufficient to build a standalone trading strategy around. The signal-to-noise ratio in daily returns is a bit too high for any single NLP-derived feature to dominate.

Extensions. Three natural next steps: (1) use blame as one signal among many in a multi-factor trading strategy, (2) test whether the effect is stronger under specific conditions (e.g., firms that miss earnings *and* blame-shift, or firms with a history of blame-shifting), and (3) explore whether the market reaction to blame-shifting has changed over time as NLP tools have become more widespread.

10. Challenges Faced

Scoring methodology. We initially computed the industry-relative blame score by dividing by the industry average. This created extreme values when the denominator was near zero. Switching to subtraction (firm change minus industry average change) produced a cleaner, more interpretable signal.

Earnings surprise scaling. Our first attempt scaled earnings surprise by price (EPS / price) or by actual EPS. Both introduced distortions for firms with small denominators. We ultimately used the raw IBES surprise (actual minus estimate) directly, which is simpler and avoids these problems.

Probability vs. binary labels. Initially, FinBERT classified each sentence as blame (1) or not (0). We found that using the continuous probability of the positive class (e.g., 0.73 instead of 1) preserved much more signal. Hard cutoffs throw away the model's uncertainty, which carries information.

Fama-MacBeth correction. Our original regression pooled all observations and treated them as independent. This violates a basic OLS assumption: earnings calls cluster in time, so errors within a quarter are correlated. Pooled standard errors were too small. Switching to Fama-MacBeth, which treats each quarter as one independent unit, was the correct fix.